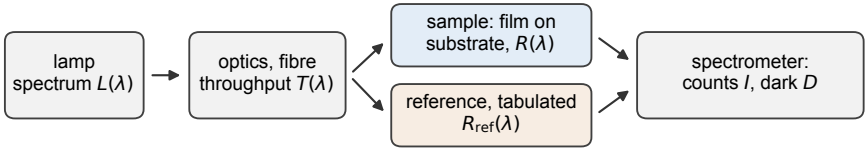

$$\widehat{R}(\lambda) = R_{\mathrm{ref}}(\lambda)\,\frac{I_{\mathrm{s}}(\lambda) - D_{\mathrm{s}}(\lambda)}{I_{\mathrm{r}}(\lambda) - D_{\mathrm{r}}(\lambda)}$$

lamp or throughput drift between the two measurements, uniform in λ	*scalar gain g (this paper)*
error in the tabulated R_{ref}, or drift that varies with λ	*spectral gain $g(\lambda)$*
stray light, incomplete dark subtraction	*additive offset*
detector response shared by both measurements (linear, stable)	*cancels in the ratio*